

Multi PCSM

IEC

Easy maintenance.

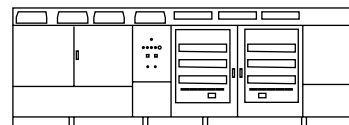
Integrated MV solution in the same enclosure.

Up to 4 independent DC inputs.

Advanced grid support.

Compatible with all battery technologies.

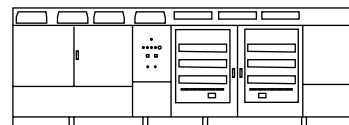




REFERENCES		FP5150MH2	FP5150MH4
AC	AC Output Power (kVA/kW) @30 °C ^[1]	5360	
	AC Output Power (kVA/kW) @35 °C ^[1]	5150	
	AC Output Power (kVA/kW) @40 °C ^[1]	4940	
	AC Output Power (kVA/kW) @50 °C ^[1]	4520	
	Operating Grid Voltage (kV)	34.5 kV ±10%	
	Rated Impulse Withstand Voltage	145 kV	
	Operating Grid Frequency (Hz)	60 Hz	
	Current Harmonic Distortion (THDi)	< 3% per IEEE 519	
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging	
	Reactive Power Compensation	Four quadrant operation	
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s	
DC	DC Voltage Range Full Power ^[3]	1019 V - 1500 V	
	Maximum DC Voltage	1500 V	
	DC Voltage Ripple	< 3%	
	Max. DC Continuous Current (A)	2684	1342
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms	
	Battery Technology	All type of batteries (BMS required)	
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η)	98.00% including MV transformer	
	CEC (η)	97.53% including MV transformer	
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5	
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3	
	Weight (lbs)	30865	
	Weight (kg)	14000	
	Type of Ventilation	Forced air cooling	
ENVIRONMENT	Degree of Protection	IP55	
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >30 °C power derating	
	Operating Relative Humidity Range	From 4% to 100%	
	Storage Temperature Range	From -40 °C to +60 °C	
	Max. Altitude (above sea level) ^[5]	2000 m	
CONTROL	Communication Protocol	Modbus TCP	
	Grid Forming (VISMA) / Black Start Capable	Standard	
PROTECTIONS	Ground Fault Protection	Insulation monitoring device	
	Humidity Control	Active heating	
	General AC Protection & Disconnection	36 kV MV switchgear (2L+V)	
	General DC Protection & Disconnection	High-speed fuses, Motorized DC disconnect switches ^[6]	
	Overvoltage Protection	Type 2 for AC and Type 1+2 for DC	
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-1 / IEC 62477-2	
	Utility Interconnect ^[7]	IEC 62116 / / G99 / VDE 4110-4120-4130 / CEI 0-16 / NTS 2.1 / EN 50549	

NOTES

- [1] Values at 1.00-Vac nom and CosPhi=1. Consult Power Electronics for derating curves.
[2] Consult P-Q charts available: $Q(kVar) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
[3] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
[4] Optional available for temperatures below -25 °C.
[5] Consult Power Electronics for altitudes above 1000 m.
[6] Battery short circuit disconnection must be done on the battery side.
[7] Consult Power Electronics for other applicable standards / grid codes.

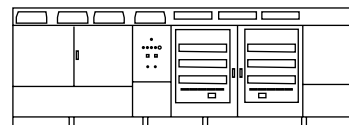


REFERENCES		FP5151MH2	FP5151MH4
AC	AC Output Power (kVA/kW) @30 °C ^[1]	5360	
	AC Output Power (kVA/kW) @35 °C ^[1]	5150	
	AC Output Power (kVA/kW) @40 °C ^[1]	4940	
	AC Output Power (kVA/kW) @50 °C ^[1]	4520	
	Operating Grid Voltage (kV)	33 kV ±10%	
	Rated Impulse Withstand Voltage	145 kV	
	Operating Grid Frequency (Hz)	50 Hz	
	Current Harmonic Distortion (THDi)	< 3% per IEEE 519	
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging	
	Reactive Power Compensation	Four quadrant operation	
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s	
DC	DC Voltage Range Full Power ^[3]	1019 V - 1500 V	
	Maximum DC Voltage	1500 V	
	DC Voltage Ripple	< 3%	
	Max. DC Continuous Current (A)	2684	1342
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms	
	Battery Technology	All type of batteries (BMS required)	
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η)	98.00% including MV transformer	
	CEC (η)	97.53% including MV transformer	
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5	
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3	
	Weight (lbs)	30865	
	Weight (kg)	14000	
	Type of Ventilation	Forced air cooling	
ENVIRONMENT	Degree of Protection	IP55	
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >30 °C power derating	
	Operating Relative Humidity Range	From 4% to 100%	
	Storage Temperature Range	From -40 °C to +60 °C	
	Max. Altitude (above sea level)	1000 m	
CONTROL	Communication Protocol	Modbus TCP	
	Grid Forming (VISMA) / Black Start Capable	Standard	
PROTECTIONS	Ground Fault Protection	Insulation monitoring device	
	Humidity Control	Active heating	
	General AC Protection & Disconnection	36 kV MV switchgear (2L+V)	
	General DC Protection & Disconnection	High-speed fuses, Motorized DC disconnect switches ^[5]	
	Overvoltage Protection	Type 2 for AC and Type 1+2 for DC	
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-1 / IEC 62477-2	
	Utility Interconnect ^[6]	IEC 62116 / / G99 / VDE 4110-4120-4130 / CEI 0-16 / NTS 2.1 / EN 50549	

NOTES

- [1] Values at 1.00·Vac nom and CosPhi=1. Consult Power Electronics for derating curves.
[2] Consult P-Q charts available: $Q(kVar) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
[3] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
[4] Optional available for temperatures below -25 °C.
[5] Battery short circuit disconnection must be done on the battery side.
[6] Consult Power Electronics for other applicable standards / grid codes.

The content of this document is periodically updated. Power Electronics reserve the right to modify all or part of the contents of this document without previous notice.



REFERENCES	FP5152MH2	FP5152MH4
AC	AC Output Power (kVA/kW) @30 °C ^[1]	5360
	AC Output Power (kVA/kW) @35 °C ^[1]	5150
	AC Output Power (kVA/kW) @40 °C ^[1]	4940
	AC Output Power (kVA/kW) @50 °C ^[1]	4520
	Operating Grid Voltage (kV)	30 kV ±10%
	Rated Impulse Withstand Voltage	145 kV
	Operating Grid Frequency (Hz)	50 Hz
	Current Harmonic Distortion (THDi)	<3% per IEEE 519
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging
	Reactive Power Compensation	Four quadrant operation
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s
DC	DC Voltage Range Full Power ^[3]	1019 V - 1500 V
	Maximum DC Voltage	1500 V
	DC Voltage Ripple	<3%
	Max. DC Continuous Current (A)	2684
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms
	Battery Technology	All type of batteries (BMS required)
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η)	98.00% including MV transformer
	CEC (η)	97.53% including MV transformer
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5
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ENVIRONMENT	Degree of Protection	IP55
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >30 °C power derating
	Operating Relative Humidity Range	From 4% to 100%
	Storage Temperature Range	From -40 °C to +60 °C
	Max. Altitude (above sea level)	1000 m
CONTROL	Communication Protocol	Modbus TCP
	Grid Forming (VISMA) / Black Start Capable	Standard
PROTECTIONS	Ground Fault Protection	Insulation monitoring device
	Humidity Control	Active heating
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